

Fixpoint Operators

Operator	Semantics
<code>nu X. phi(X)</code>	Greatest fixpoint — starts from <i>all states</i> and removes violators downward. Captures safety and invariance .
<code>mu X. phi(X)</code>	Least fixpoint — starts from <i>no states</i> and adds satisfiers upward. Captures reachability and liveness .

Intuition: `nu` keeps states that *always* satisfy φ ; `mu` finds states that *eventually* satisfy φ .
Fixpoint variables (`X`) are recursive: the formula “unfolds” until a stable set is reached.

Modalities

Syntax	Meaning
<code>[] phi</code>	Box: <i>all</i> successors satisfy φ
<code><> phi</code>	Diamond: <i>some</i> successor satisfies φ

Without labels, modalities quantify over **all** outgoing transitions from the current state.

Boolean Operators

<code>phi && psi</code>	Conjunction (and)
<code>phi psi</code>	Disjunction (or)
<code>! phi</code>	Negation (not)
<code>true</code>	Holds in every state
<code>false</code>	Holds in no state

Labeled Modalities

Restrict modalities to specific labels:

```
// All a-or-b transitions lead to phi
[ labels = {a, b} ] phi

// Some a-transition leads to phi
< labels = {a} > phi
```

- Without `labels = {...}`: considers **all** transitions.
- With labels: considers **only** transitions on the specified labels.
- Multiple labels in the set act as a union.

Nesting and Alternation

Formulas can nest fixpoints and modalities:

```
// Alternation: nu outside, mu inside
nu X. (mu Y. (p || <> Y)) && ([] X)
```

- Alternation depth** affects complexity — deeper nesting is more expensive.
- Inner `mu` resets each time the outer `nu` iterates.
- Most practical properties have alternation depth ≤ 2 .

Common Patterns

Pattern	Formula
Safety invariant	<code>nu X. ([] X)</code>
Reachability	<code>mu X. (target <> X)</code>
GF (inf. often)	<code>nu X. (mu Y. (p <> Y)) && ([] X)</code>
No deadlock	<code>nu X. (<> true) && ([] X)</code>
Labeled check	<code>(! src) [labels = {a}] dst</code>
Mutual exclusion	<code>(! A) (! B)</code>
Recovery	<code>(! bad) (mu Y. (good <> Y))</code>
Ctrl reachability	<code>mu X. (p [(ctrl = controllable)] X)</code>
Env diamond	<code>< (ctrl = environment) > phi</code> — some uncontrollable satisfies OR all controllable satisfy

Reading Patterns

- Safety** = “always stays in good states” $\Rightarrow$ `nu` + `[]`
- Reachability** = “eventually reaches target” $\Rightarrow$ `mu` + `<>`
- Liveness** = “always eventually” $\Rightarrow$ `nu` wrapping `mu`
- Implications ($p \Rightarrow q$) encoded as `(! p) || q`

Common Mistakes

- Swapping `nu/mu` — reverses safety vs. reachability semantics.
- Using `<>` (diamond) when `[]` (box) is needed — “some path” vs. “all paths”.
- Forgetting that `[] false` holds vacuously in deadlock states (no successors).

Further Reading

Kozen (1983) “Results on the Propositional Mu-Calculus” — fixpoint semantics. Emerson & Jutla (1991) — positional determinacy of parity games. Bruse, Friedmann & Lange (2016) — strategy extraction from fixpoint iteration.